Problem 2: discussion questions

1. It includes learning blocks such as:
    a. Classification: IRIS flower classifier using tabular features
    b. Regression: predicting a continuous value such as temperature, vibration level, or battery voltage
    c. Image Classification / Transfer Learning: mask vs no-mask image classifier
    d. Keyword Spotting / Audio Classification: detecting spoken words such as “yes,” “no,” or “silence”
    e. Object Detection / FOMO: detecting and locating objects in images, such as people or small objects
    f. Classical ML: feature-based classification using simpler machine learning models
    g. Anomaly Detection, K-means or GMM: fan or machine fault detection
    h. Custom Learning Block: a user-defined Keras, PyTorch, or scikit-learn model for a special task
2. Common layer types include:
    a. Dense: fully connected layers used to learn class decision boundaries.
    b. Reshape: changes the input shape without changing the data values
    c. Flatten: converts multidimensional features into a 1D vector before feeding them into Dense layers
    d. Dropout: randomly disables some neurons during training to reduce overfitting
    e. 1D Convolution: useful for sequential data such as audio or time-series sensor data
    f. 2D Convolution: useful for image-like inputs such as spectrograms or camera images
    g. Pooling (1D/2D): reduce feature-map size and help lower computation
    h. Softmax output layer: for classification, so the model outputs class probabilities
    i. Input layer: receives the processed features and passes them to subsequent layers
3. The compression options include:
    a. Unoptimized: exports the model without extra compression and is useful as a baseline for comparing size, RAM, and accuracy
    b. INT8 quantization: compresses the model by converting 32-bit floating-point weights/operations to 8-bit integer format, which reduces memory and can improve speed/power efficiency.

   c. EON Compiler: further optimizes the model for edge devices by reducing memory and computation requirements
4. Edge Impulse synthetic data supports:
   a. Image: the available generation blocks include DALL-E Image Generation and FLUX Pro Image Generation, which generate image samples from text prompts
   b. Audio: the Whisper Keyword Spotting Generation block can generate speech/keyword samples from text, which is useful for keyword spotting tasks
   c. Time-series data: Physics Simulations utilize 3D digital twins and simulation tools to generate artificial sensors and accelerometer data mimicking real-world conditions.

   Synthetic data is important because it can increase dataset size and diversity when real data is limited, expensive, or hard to collect. It can also help cover rare cases, improve model robustness, and reduce privacy concerns because the generated data does not have to come directly from real users.